

Koushik Deb

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Education

East Point College of Engineering and Technology

Bachelor of Technology in Computer Science Engineering, CGPA – 8.9/10

August 2023 – June 2027*

Bengaluru, Karnataka

Coursework: Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks, Object Oriented Programming

Certifications

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional

Issued Oct 2025

Oracle AI Vector Search Certified Professional

Issued Oct 2025

Oracle Cloud Infrastructure 2025 AI Foundations Associate

Issued Oct 2025

Data Analytics Job Simulation – Deloitte

Issued Mar 2026

Java Foundations – Oracle Academy

Issued Nov 2024

Projects

PulseKin – Federated AI Healthcare Platform

February 2026

Next.js, TypeScript, FastAPI, Flask, XGBoost, PyTorch, SHAP, Groq LLM

- Engineered a full-stack healthcare platform with 3 role-based dashboards (Patient, Doctor, Admin) using Next.js 16, React 19, and TypeScript, deployed on Vercel with 2 backend microservices on Railway.
- Developed 4 XGBoost classifiers for disease risk prediction (Diabetes, Heart, Kidney, Liver) using SMOTE balancing and stratified cross-validation, delivering real-time diagnostic scores to doctors.
- Integrated SHAP via TreeExplainer for per-prediction feature attribution, enabling doctors to visualize which biomarkers drive each patient's risk score with explainable AI.
- Built a 4-layer PyTorch Deep Neural Network with BatchNormalization and Dropout regularization for biomarker risk scoring across 8 clinical features.
- Implemented conversational AI intake assistant powered by Groq Llama 3.1, collecting patient symptoms through multi-turn structured prompting and auto-generating clinical PDF reports via ReportLab.
- Developed Prescription Vision OCR pipeline using Groq multimodal model to parse prescription images into structured medicine arrays with zero-persistence privacy architecture.

SafeGuard – AI Women Safety App

December 2025

React Native, Expo, TypeScript, Express.js, PostgreSQL, Drizzle ORM, Gemini 2.5 Flash

- Built a cross-platform mobile safety application with 5 tab-based navigation screens using Expo and React Native, integrating Google Gemini 2.5 Flash for real-time threat detection.
- Implemented multi-signal threat detection combining AI audio analysis (5-second clips every 8 seconds), multilingual distress keyword spotting across 6 languages, and accelerometer-based motion anomaly detection.
- Developed REST API server using Express 5 with TypeScript, featuring real-time GPS tracking, SOS alert management, emergency contacts CRUD, and WebSocket audio streaming.
- Engineered PostgreSQL database schema with Drizzle ORM for profiles, emergency contacts, SOS alerts with trigger classification, and timestamped GPS location tracking.

Veritas – Misinformation Detection Platform

June 2025

React Native, Expo, TypeScript, Python, Flask, AI/ML

- Developed a mobile application for real-time misinformation detection using React Native with Expo frontend paired with a Python Flask AI classification backend deployed on Render.
- Implemented modular microservices architecture with separate frontend, backend API server, and ML classification service for scalable news credibility analysis.

Global Crisis Tracker

May 2025

HTML, CSS, JavaScript, News API, OpenWeather API

- Built a dynamic web platform providing real-time updates on global crises with interactive dashboard featuring maps, charts, and region-based breakdowns.
- Integrated News API and OpenWeather API for live data feeds with responsive design and dark mode toggle, enhancing cross-device user experience.

- Developed a ticket booking system using C with linked lists and queues for real-time seat allocation and queue management, optimizing booking operations and system performance.
- ISL Translator Chrome Extension (Smart India Hackathon)

2024
- HTML, CSS, JavaScript, Web APIs
- Built a Chrome extension using webcam input to convert Indian Sign Language gestures into real-time text and speech output for the deaf and hard-of-hearing community.

Technical Skills

Languages: JavaScript, TypeScript, Python, Java, C
Frontend Tools: React.js, Next.js, HTML5, CSS3, Tailwind CSS, Bootstrap, Responsive Web Design
Backend Tools: Node.js, Express.js, FastAPI, Flask, REST API Design, WebSocket
Databases: MongoDB, MySQL, PostgreSQL, Drizzle ORM
AI/ML Tools: XGBoost, PyTorch, SHAP, SMOTE, Scikit-learn, Groq LLM (Llama 3.1), Google Gemini 2.5 Flash
Production Tools: Docker, Git, GitHub, Vercel, Railway, Render, Cloudinary, React Native, Expo

Achievements

- Participation at 5+ Hackathons across Bengaluru and National Level

2024 – Present
- Participated as Frontend Developer in Smart India Hackathon (National Level), building an ISL Translator Extension for accessibility.
 - Secured 1st Runner-Up at College Code Battle, solving complex algorithmic challenges under timed constraints using DSA concepts.
 - Competed at InnovateX Hackathon, Presidency University, Bengaluru, collaborating on rapid prototyping and innovative technology solutions.
 - Competed at Zynex Hackathon, Cambridge Institute of Technology, Bengaluru, building solutions under time constraints with cross-functional teams.
 - Competed at HackSphere Hackathon, TJohn Institute of Technology, Bengaluru, demonstrating full-stack development and collaborative delivery.